

GNN_Model

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seed(46)
Building KNN graphs (k=5)...
  Building train graphs...
  Building test graphs...
X_train shape : (173066, 16, 20)
ADJ_train shape: (173066, 16, 16)
X_test shape : (19230, 16, 20)
Test data saved to output2/gnn_tagger_v1
Checking sample_weight:
[2.32066617 0.99185044 0.32742429 ... 0.32742429 0.65398742 2.32066617]
GNN Input shape (16, 20)
Model: "GNNModel"
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Layer (type)                Output Shape              Param #
Connected to
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model_input (InputLayer)    [(None, 16, 20)]         0          []
bn_input (BatchNormalizat   (None, 16, 20)           80
['model_input[0][0]']
ion)
adj_input (InputLayer)      [(None, 16, 16)]         0          []
gcn_dense_0 (QDense)        (None, 16, 32)           672
['bn_input[0][0]']
gcn_agg_0 (Dot)             (None, 16, 32)           0
['adj_input[0][0]',
'gcn_dense_0[0][0]']
gcn_act_0 (QActivation)     (None, 16, 32)           0
['gcn_agg_0[0][0]']
gcn_dense_1 (QDense)        (None, 16, 32)           1056
['gcn_act_0[0][0]']
gcn_agg_1 (Dot)             (None, 16, 32)           0
['adj_input[0][0]',
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'gcn_dense_1[0][0]']

gcn_act_1 (QActivation)      (None, 16, 32)      0
['gcn_agg_1[0][0]']

act_pool (QActivation)      (None, 16, 32)      0
['gcn_act_1[0][0]']

pool (GlobalAveragePooling  (None, 32)          0
['act_pool[0][0]']
1D)

dense_0 (QDense)            (None, 32)          1056
['pool[0][0]']

act_0 (QActivation)         (None, 32)          0
['dense_0[0][0]']

dense_1 (QDense)            (None, 16)          528
['act_0[0][0]']

act_1 (QActivation)         (None, 16)          0
['dense_1[0][0]']

jet_pt_regression_dense (Q  (None, 8)           136
['act_1[0][0]']
Dense)

jet_id_classification (QDe  (None, 8)           136
['act_1[0][0]']
nse)

jet_pt_regression_act (QAc  (None, 8)           0
['jet_pt_regression_dense[0][0]
tivation)

jet_id_output (Softmax)     (None, 8)           0
['jet_id_classification[0][0]']

pT_output (QDense)         (None, 1)           9
['jet_pt_regression_act[0][0]']

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Total params: 3673 (14.35 KB)
Trainable params: 3633 (14.19 KB)
Non-trainable params: 40 (160.00 Byte)

Epoch 10/10

153/153 - 3s -

loss: 1.2520 -

jet_id_output_loss: 1.1973 -

pT_output_loss: 0.0547 -

jet_id_output_categorical_accuracy: 0.5337 -

jet_id_output_weighted_categorical_accuracy: 0.5414 -

pT_output_mae: 0.2493 -

pT_output_mean_squared_error: 0.1147 -

pT_output_weighted_mae: 0.2379 -

pT_output_weighted_mean_squared_error: 0.1093 -

val_loss: 1.2347 -

val_jet_id_output_loss: 1.1821 -

val_pT_output_loss: 0.0526 -

val_jet_id_output_categorical_accuracy: 0.5423 -

val_jet_id_output_weighted_categorical_accuracy: 0.5474 -

val_pT_output_mae: 0.2512 -

val_pT_output_mean_squared_error: 0.1143 -

val_pT_output_weighted_mae: 0.2346 -

val_pT_output_weighted_mean_squared_error: 0.1054 -

lr: 0.0010 - 3s/epoch - 17ms/step



